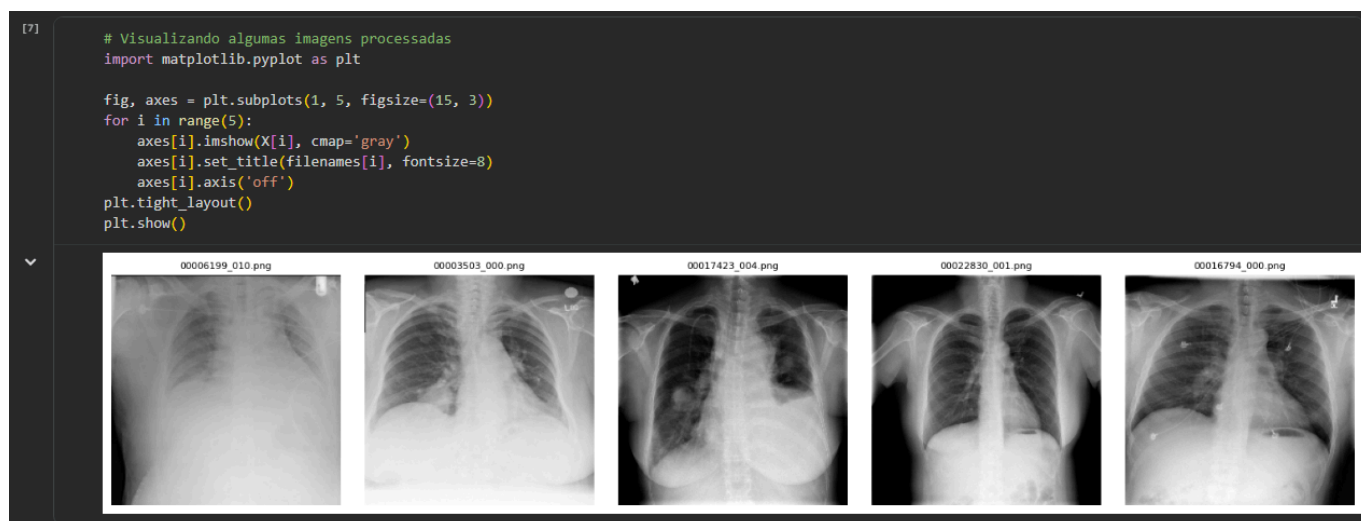
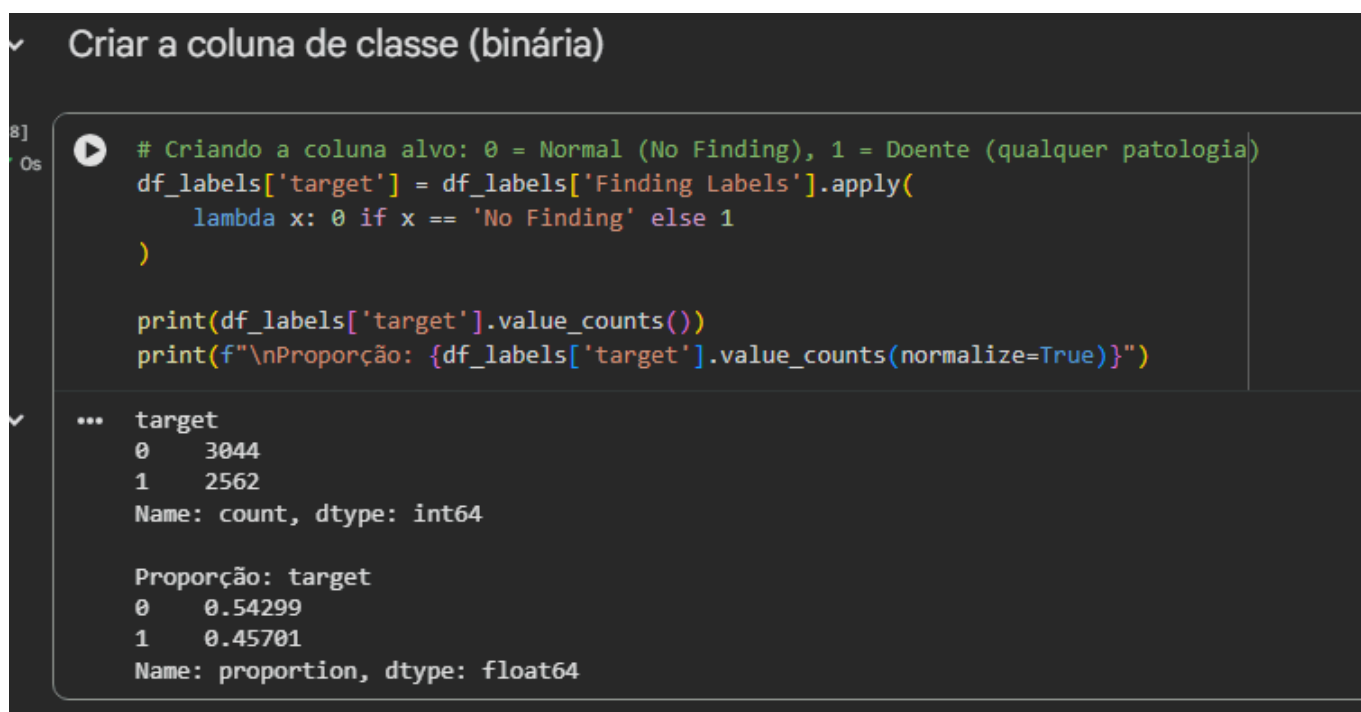


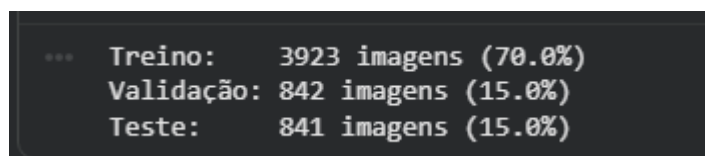
Prints das métricas de avaliação.



"Figura 1 – Visualização de amostras do dataset NIH Chest X-rays após pré-processamento (redimensionamento 224×224 e normalização)"



"Figura 2 – Distribuição das classes no dataset: 3.044 imagens normais (54,3%) e 2.562 doentes (45,7%)"



"Figura 3 – Divisão do dataset em conjuntos de treino (70%), validação (15%) e teste (15%), com stratify para manter proporção das classes"

[11]

```
# Verificando a proporção de classes
import pandas as pd

for nome, y_set in [('Treino', y_train), ('Validação', y_val), ('Teste', y_test)]:
    proporcao = np.bincount(y_set) / len(y_set)
    print(f"{nome}: Normal={proporcao[0]:.2%}, Doente={proporcao[1]:.2%}")
```

Treino: Normal=54.30%, Doente=45.70%
 Validação: Normal=54.28%, Doente=45.72%
 Teste: Normal=54.34%, Doente=45.66%

"Figura 4 – Histórico de treinamento da CNN do zero ao longo de 13 épocas com Early Stopping, e curvas de acurácia e loss de treino vs. validação"

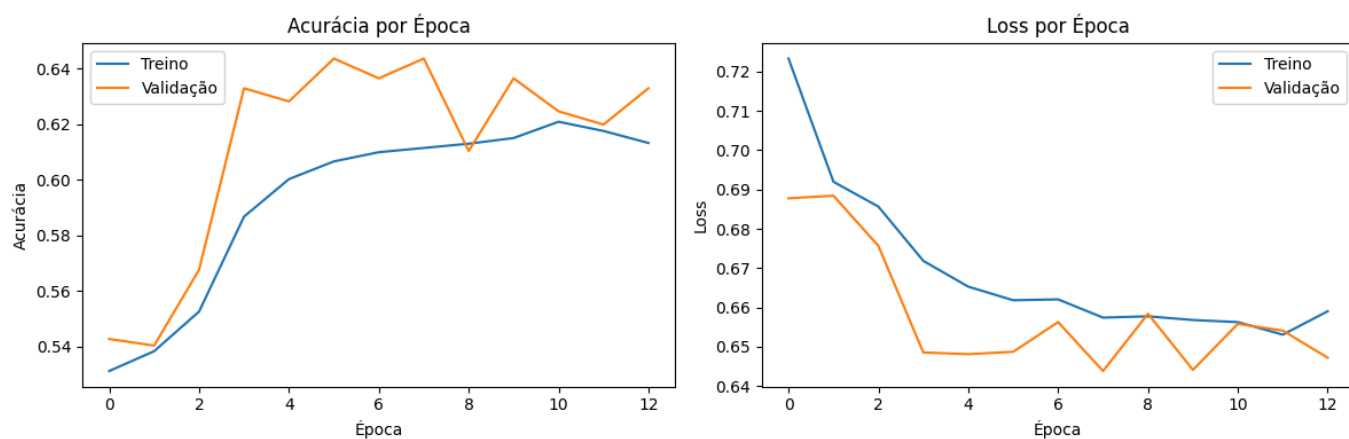
Resultados no conjunto de teste (Figuras 5, 6, 7 e 8):

[18]

✓ 4min

```
history_cnn = model_cnn.fit(
    train_generator,
    validation_data=val_generator,
    epochs=15, # reduzido de 15 -> 10 (early stopping pode parar antes mesmo disso)
    callbacks=[early_stop]
)
```

```
... Epoch 1/15
123/123 ----- 31s 186ms/step - accuracy: 0.5312 - loss: 0.7233 - val_accuracy: 0.5428 - val_loss: 0.6878
Epoch 2/15
123/123 ----- 17s 138ms/step - accuracy: 0.5384 - loss: 0.6920 - val_accuracy: 0.5404 - val_loss: 0.6884
Epoch 3/15
123/123 ----- 17s 138ms/step - accuracy: 0.5526 - loss: 0.6856 - val_accuracy: 0.5677 - val_loss: 0.6757
Epoch 4/15
123/123 ----- 19s 156ms/step - accuracy: 0.5868 - loss: 0.6718 - val_accuracy: 0.6330 - val_loss: 0.6485
Epoch 5/15
123/123 ----- 17s 138ms/step - accuracy: 0.6003 - loss: 0.6653 - val_accuracy: 0.6283 - val_loss: 0.6481
Epoch 6/15
123/123 ----- 22s 178ms/step - accuracy: 0.6067 - loss: 0.6618 - val_accuracy: 0.6437 - val_loss: 0.6487
Epoch 7/15
123/123 ----- 17s 136ms/step - accuracy: 0.6100 - loss: 0.6620 - val_accuracy: 0.6366 - val_loss: 0.6563
Epoch 8/15
123/123 ----- 18s 145ms/step - accuracy: 0.6115 - loss: 0.6574 - val_accuracy: 0.6437 - val_loss: 0.6438
Epoch 9/15
123/123 ----- 17s 141ms/step - accuracy: 0.6131 - loss: 0.6577 - val_accuracy: 0.6105 - val_loss: 0.6584
Epoch 10/15
... Epoch 1/15
123/123 ----- 31s 186ms/step - accuracy: 0.5312 - loss: 0.7233 - val_accuracy: 0.5428 - val_loss: 0.6878
Epoch 2/15
123/123 ----- 17s 138ms/step - accuracy: 0.5384 - loss: 0.6920 - val_accuracy: 0.5404 - val_loss: 0.6884
Epoch 3/15
123/123 ----- 17s 138ms/step - accuracy: 0.5526 - loss: 0.6856 - val_accuracy: 0.5677 - val_loss: 0.6757
Epoch 4/15
123/123 ----- 19s 156ms/step - accuracy: 0.5868 - loss: 0.6718 - val_accuracy: 0.6330 - val_loss: 0.6485
Epoch 5/15
123/123 ----- 17s 138ms/step - accuracy: 0.6003 - loss: 0.6653 - val_accuracy: 0.6283 - val_loss: 0.6481
Epoch 6/15
123/123 ----- 22s 178ms/step - accuracy: 0.6067 - loss: 0.6618 - val_accuracy: 0.6437 - val_loss: 0.6487
Epoch 7/15
123/123 ----- 17s 136ms/step - accuracy: 0.6100 - loss: 0.6620 - val_accuracy: 0.6366 - val_loss: 0.6563
Epoch 8/15
123/123 ----- 18s 145ms/step - accuracy: 0.6115 - loss: 0.6574 - val_accuracy: 0.6437 - val_loss: 0.6438
Epoch 9/15
123/123 ----- 17s 141ms/step - accuracy: 0.6131 - loss: 0.6577 - val_accuracy: 0.6105 - val_loss: 0.6584
Epoch 10/15
123/123 ----- 17s 139ms/step - accuracy: 0.6151 - loss: 0.6568 - val_accuracy: 0.6366 - val_loss: 0.6441
Epoch 11/15
123/123 ----- 17s 138ms/step - accuracy: 0.6210 - loss: 0.6563 - val_accuracy: 0.6247 - val_loss: 0.6558
Epoch 12/15
123/123 ----- 16s 134ms/step - accuracy: 0.6176 - loss: 0.6531 - val_accuracy: 0.6200 - val_loss: 0.6541
Epoch 13/15
123/123 ----- 17s 140ms/step - accuracy: 0.6133 - loss: 0.6590 - val_accuracy: 0.6330 - val_loss: 0.6472
```

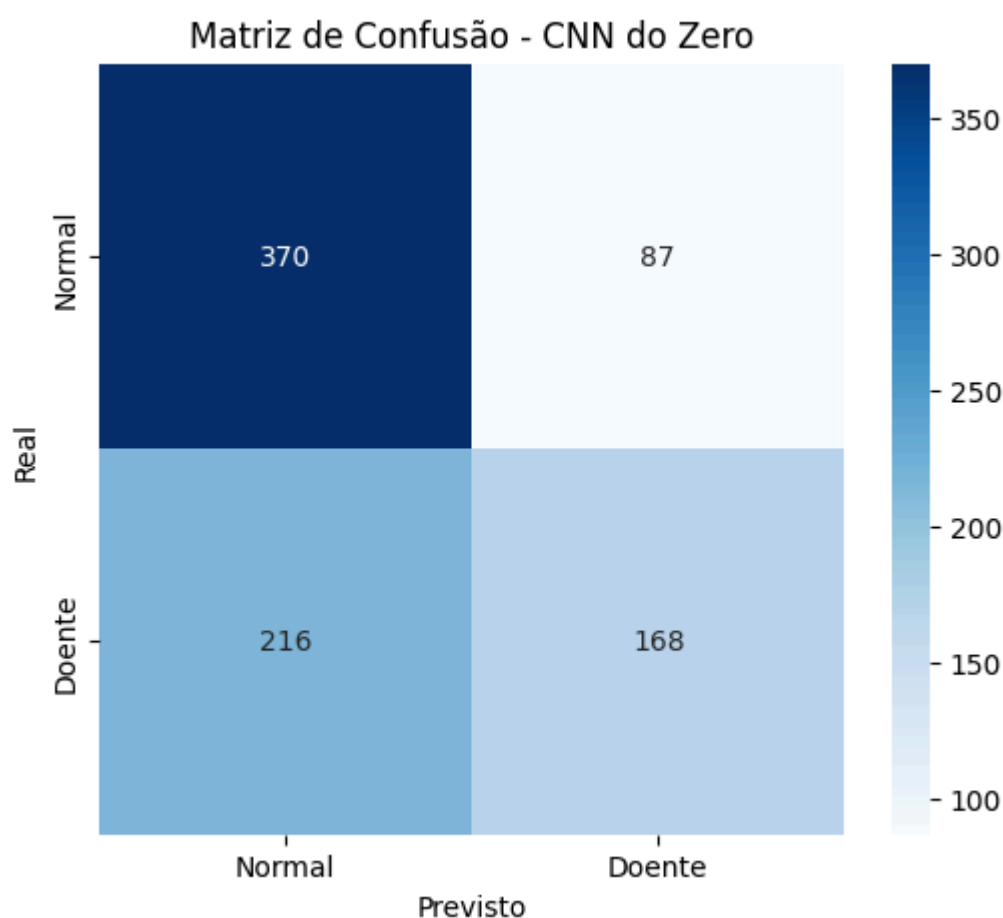


```
# Acurácia no teste

from sklearn.metrics import accuracy_score

test_accuracy = accuracy_score(y_test, y_pred)
print(f"🎯 Acurácia no conjunto de teste: {test_accuracy:.2%}")

... 🎯 Acurácia no conjunto de teste: 63.97%
```



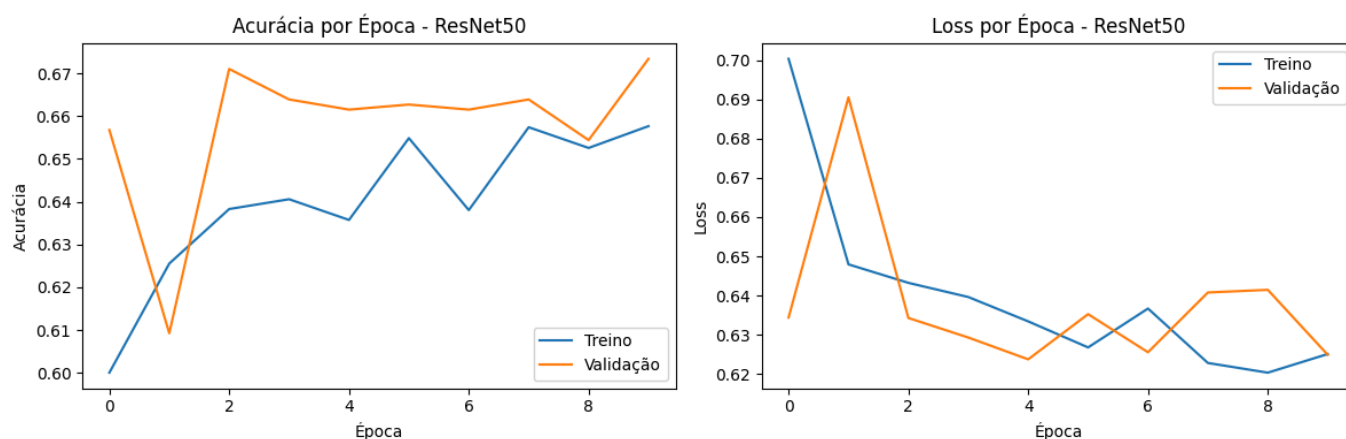
"Figura 9 – Matriz de Confusão da CNN do zero: 370 verdadeiros negativos, 168 verdadeiros positivos, 87 falsos positivos e 216 falsos negativos"

```
[31]
✓ 9min

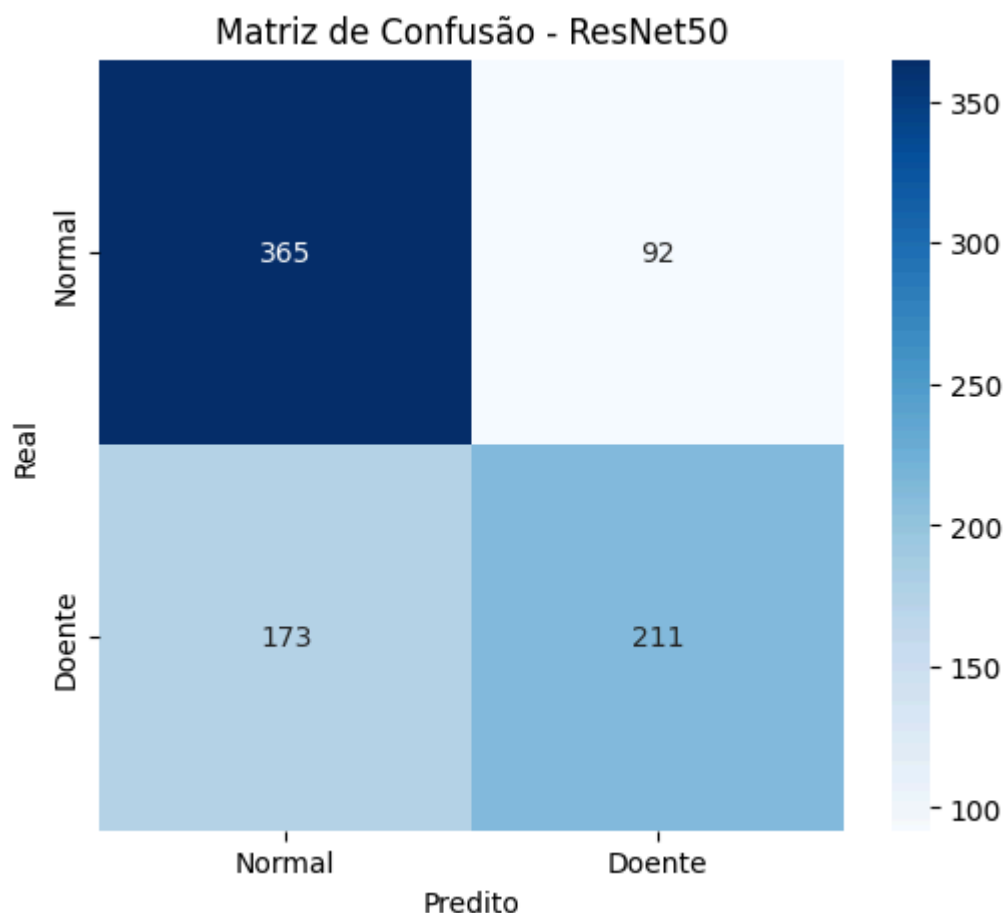
# Treinar o modelo
history_resnet = model_resnet.fit(
    train_generator_resnet,
    validation_data=val_generator_resnet,
    epochs=15,
    callbacks=[early_stop_resnet]
)

Epoch 1/15
246/246 ————— 75s 256ms/step - accuracy: 0.6001 - loss: 0.7003 - val_accuracy: 0.6568 - val_loss: 0.6344
Epoch 2/15
246/246 ————— 51s 209ms/step - accuracy: 0.6255 - loss: 0.6480 - val_accuracy: 0.6093 - val_loss: 0.6905
Epoch 3/15
246/246 ————— 51s 209ms/step - accuracy: 0.6383 - loss: 0.6433 - val_accuracy: 0.6710 - val_loss: 0.6343
Epoch 4/15
246/246 ————— 51s 208ms/step - accuracy: 0.6406 - loss: 0.6397 - val_accuracy: 0.6639 - val_loss: 0.6293
Epoch 5/15
246/246 ————— 52s 211ms/step - accuracy: 0.6357 - loss: 0.6334 - val_accuracy: 0.6615 - val_loss: 0.6238
Epoch 6/15
246/246 ————— 51s 209ms/step - accuracy: 0.6549 - loss: 0.6268 - val_accuracy: 0.6627 - val_loss: 0.6353
Epoch 7/15
246/246 ————— 52s 211ms/step - accuracy: 0.6380 - loss: 0.6367 - val_accuracy: 0.6615 - val_loss: 0.6256
Epoch 8/15
246/246 ————— 52s 210ms/step - accuracy: 0.6574 - loss: 0.6228 - val_accuracy: 0.6639 - val_loss: 0.6408
Epoch 9/15
246/246 ————— 51s 206ms/step - accuracy: 0.6526 - loss: 0.6204 - val_accuracy: 0.6544 - val_loss: 0.6415
Epoch 10/15
246/246 ————— 52s 211ms/step - accuracy: 0.6577 - loss: 0.6251 - val_accuracy: 0.6734 - val_loss: 0.6249
```

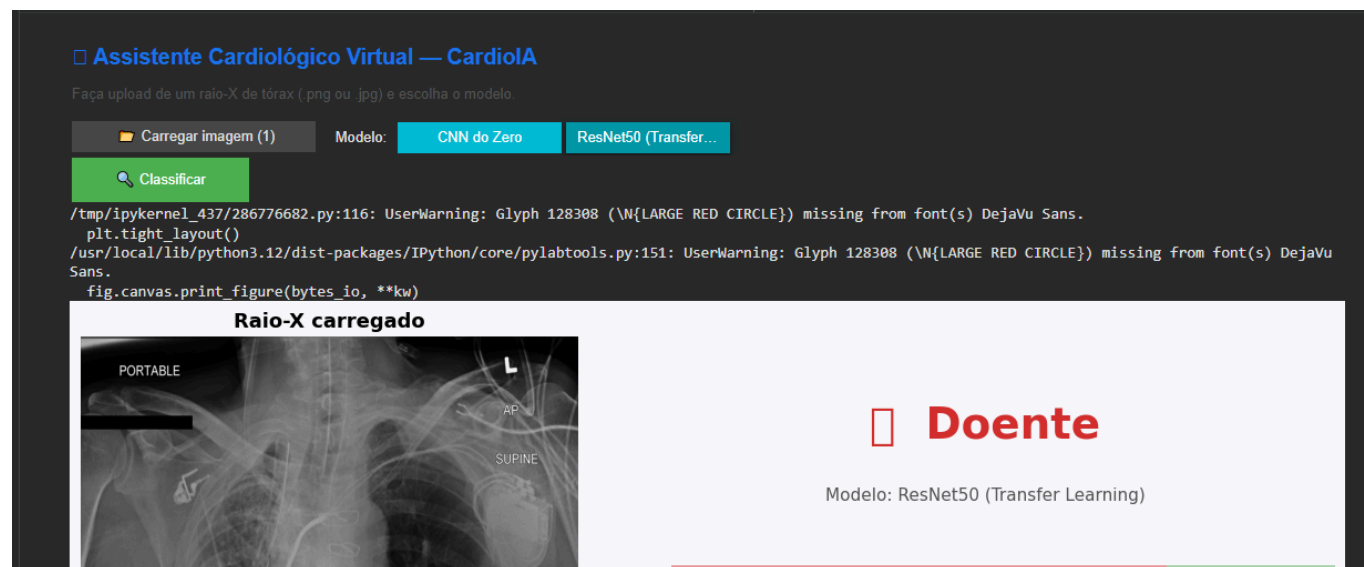
"Figura 10 – Histórico de treinamento do ResNet50 ao longo de 10 épocas com Early Stopping"



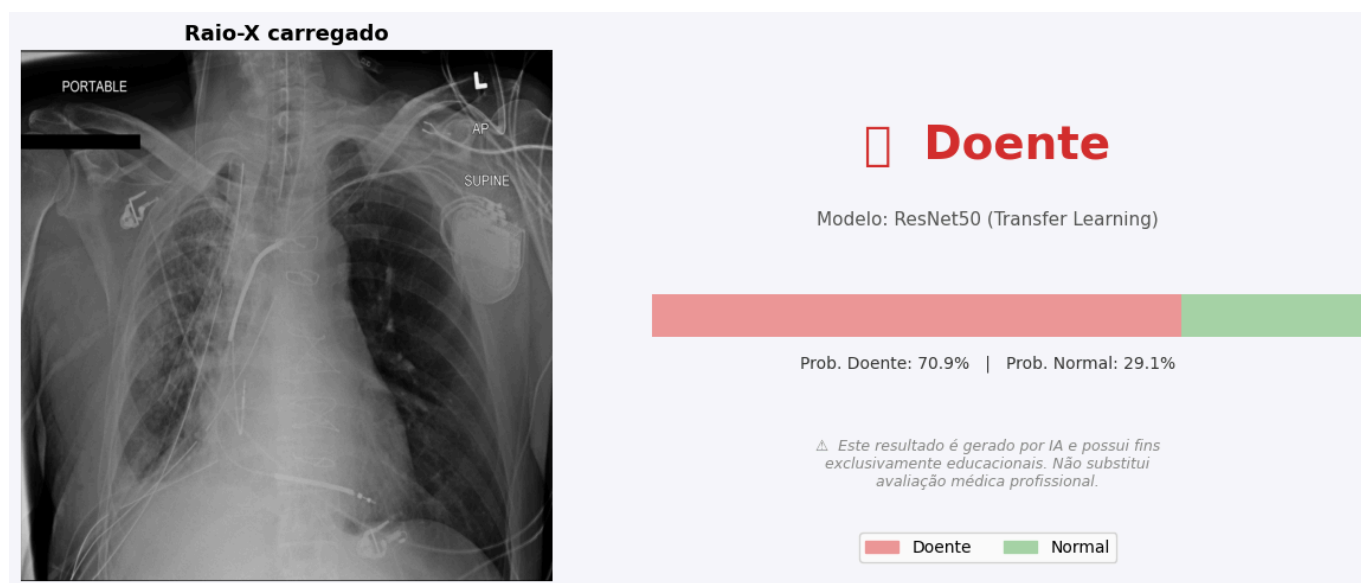
"Figura 11 – Curvas de acurácia e loss do ResNet50: treino vs. validação ao longo das épocas"



"Figura 14 – Matriz de Confusão do ResNet50: 365 verdadeiros negativos, 211 verdadeiros positivos, 92 falsos positivos e 173 falsos negativos"



"Figura 16 – Protótipo do Assistente Cardiológico Virtual em funcionamento: upload de raio-X e classificação pelo modelo ResNet50 com probabilidade de 70,9% para Doente"



"Figura 17 – Resultado completo da classificação exibindo a imagem do raio-X, diagnóstico, barra de probabilidade e aviso ético"